TEAM 83



CONTENTS

Our Solution 4

- Problem and Product Description 5
- Components 6
- Extending Mechanism and Depth Capability 7
- Engaging Mechanism at Maximum and Arbitrary Length 9
- Why choose screw threaded mechanism over other alternatives ?.....10

Flexibility and weight 11

- Material & Weight 12
- Centrifugal Retractable Blade with optimal design 13
- Blade changing Mechanism and advantages 14
- Confirmation Via Ansys Simulation.....15

Portability 16

- Achievable Compactness 17
- How to Maneuver it 17
- Dimensions 18
- Bill of Materials 19

User Friendly Interface 20

- How to use 21
- PCB Design and Electronics.....22
- MATLAB Simulink Representation 23

..

CONTENTS

Water Proofing 24

- Why Our Device is Completely Waterproof ?: Key Factors 25
- How to deal with Cavitation and corrosion .....;.....26
- Other Additional Measures 26

Design Safety from failure 27

- Determining Fatigue Factor of Safety by DE Goodman criteria.....28
- Confirmed Via FEA Simulation.....29
- Bearing Selection via SKF cataloge.....30

References 31

OUR DESIGN

SWIRL

(Stirring With Integrated Reconfigurable Lifter)

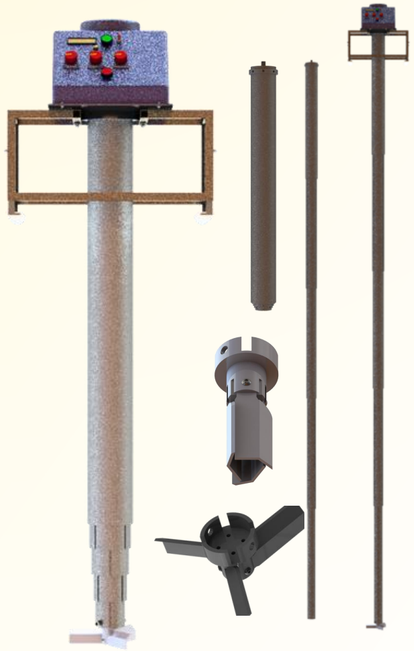
PRODUCT DESCRIPTION

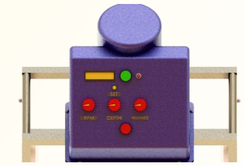
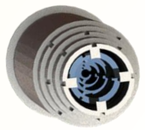
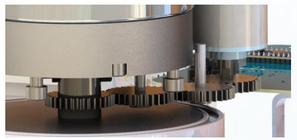
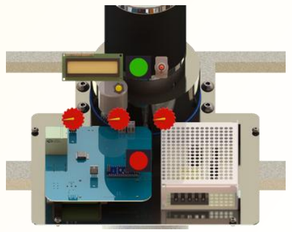
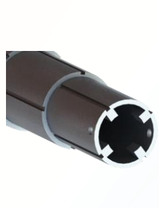
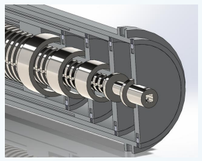
Problem Statement

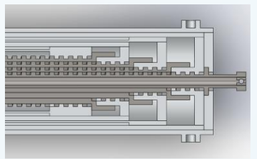
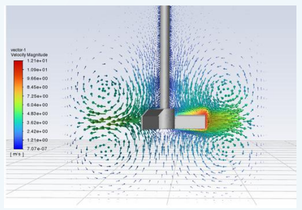
The challenge is to design a compact and innovative shaft for an agitator mechanism that can homogenize solid material or waste with liquid substances in tanks with a depth of 5 meters. The shaft needs to be retractable to a length of 1 meter for portability and extendable to 5 meters to operate in a tank of 0.4 diameter. It should be user-friendly,portable and provide waterproofing and spark-proofing for safe operation in underwater or sludge storage tanks.

Product Description

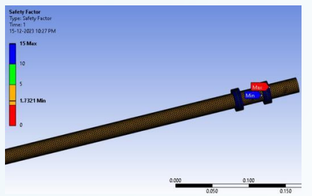
A compact slurry mixing machine designed to be portable and easily operable. This can mix thick slurry from a depth of 1m to 5m using a mechanical extending mechanism. The user interface is simple and the electronics are well packed and accomodated for a safer usage even by inexperienced operators.

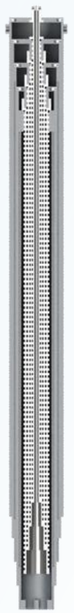


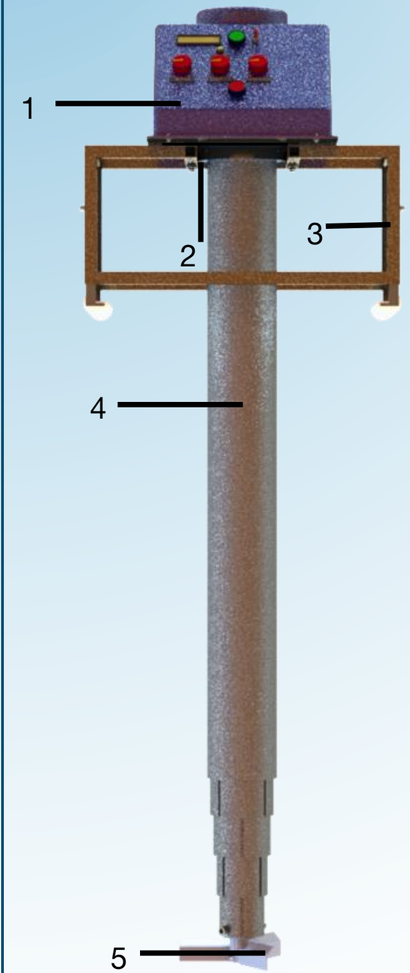



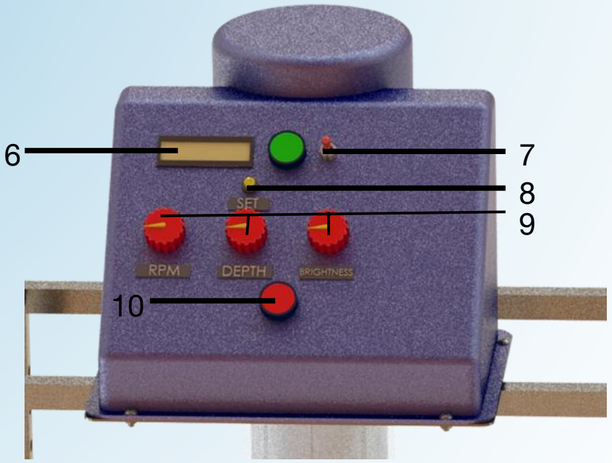


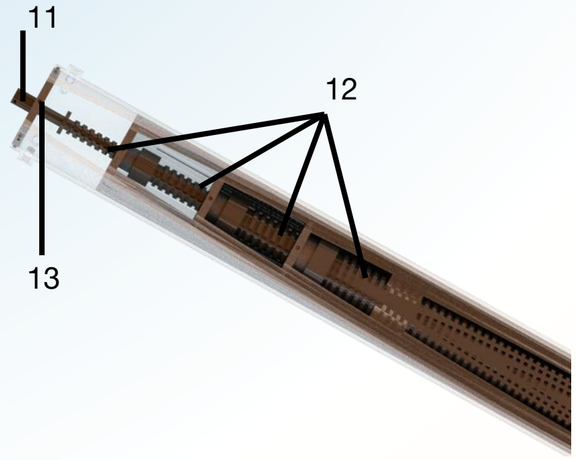


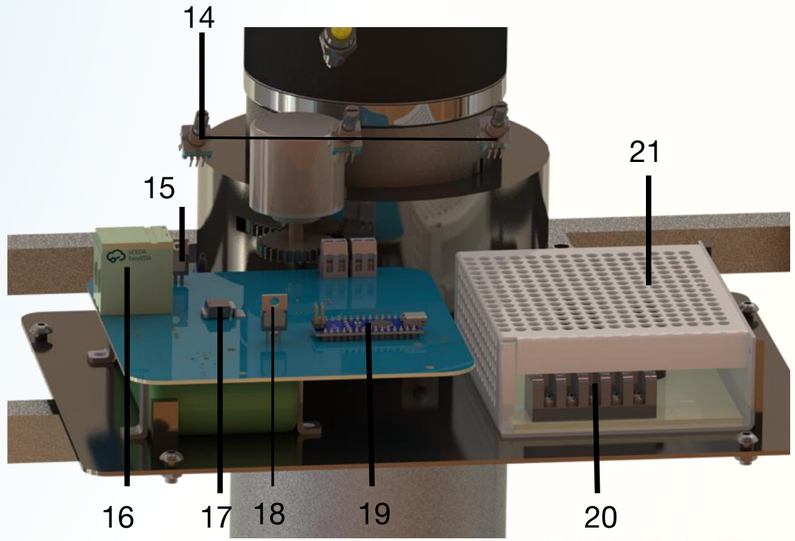


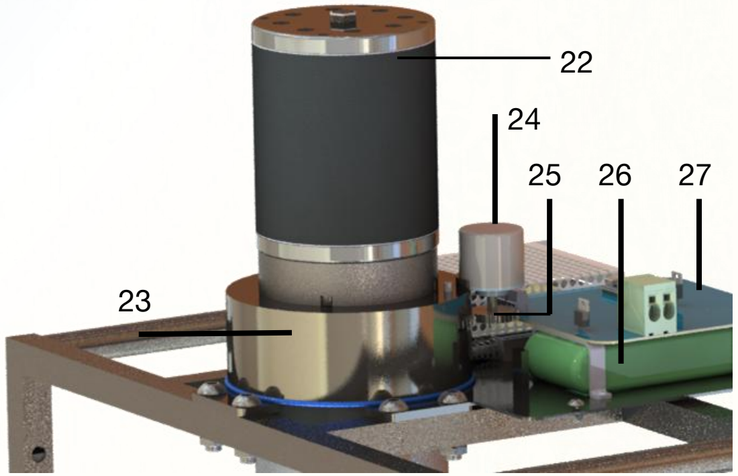
COMPONENTS





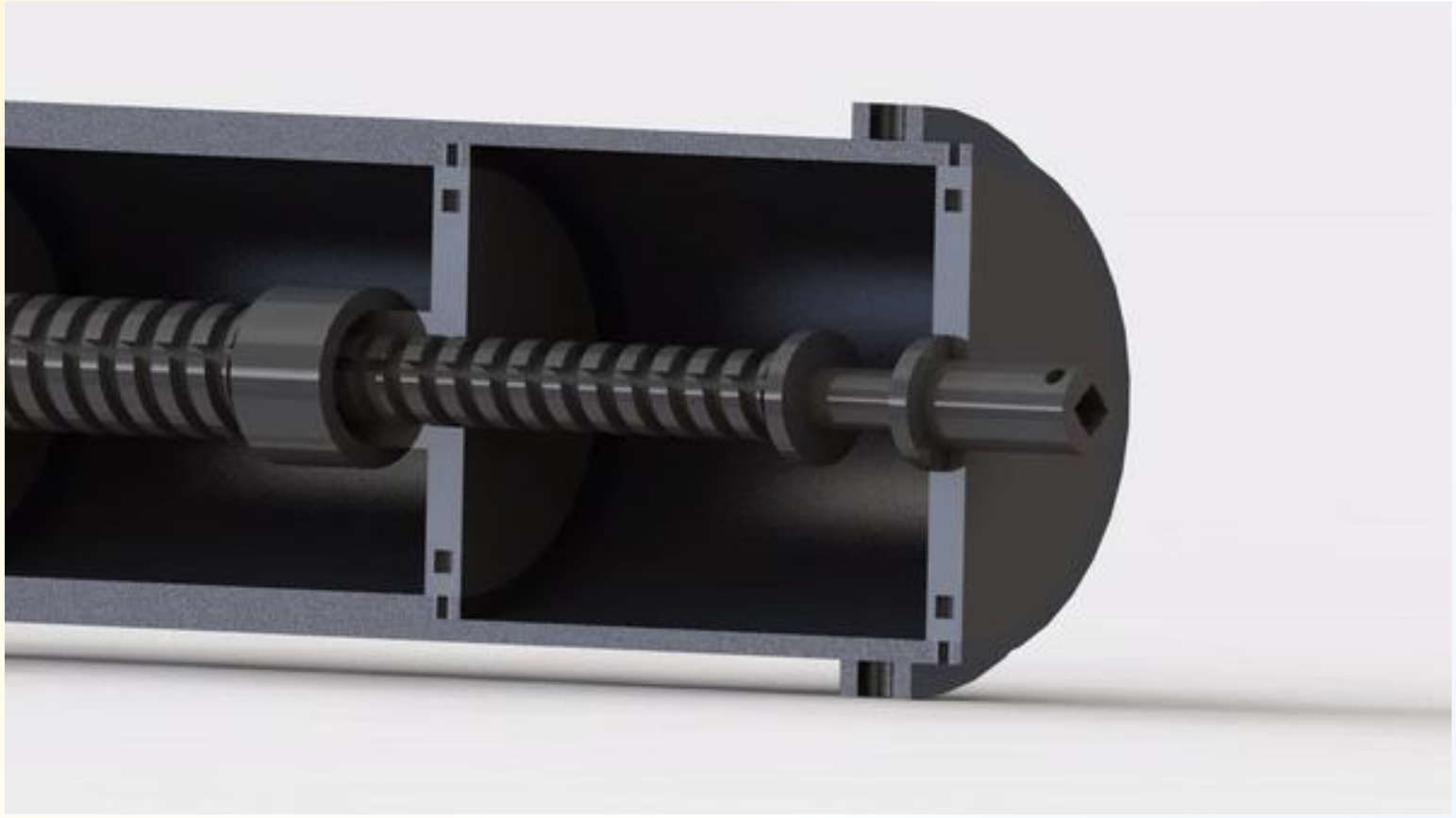


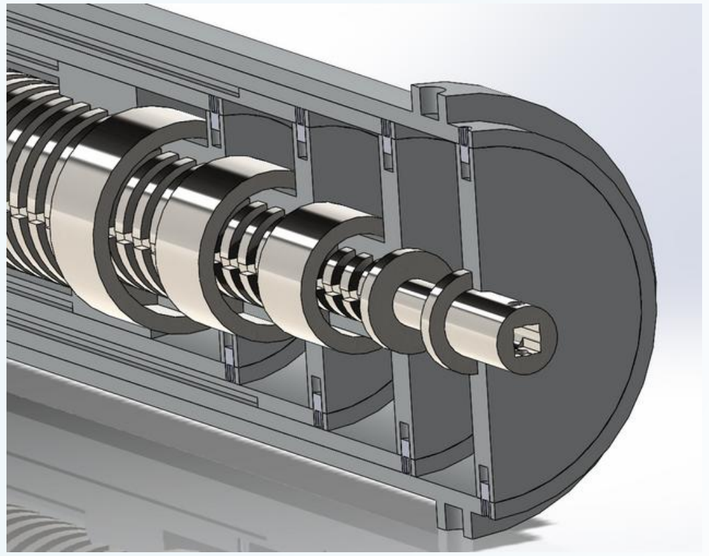




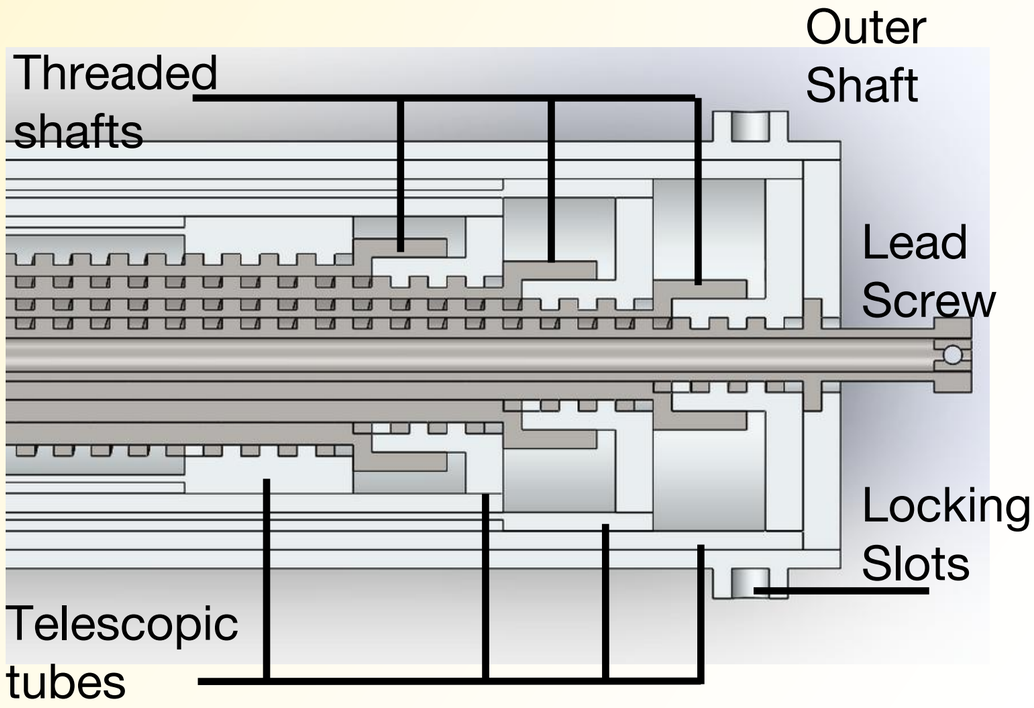
1 Central Console
2 central bearing
3 Stand
4 Main Shaft
5 Impellers
6 Display
7 Set rotation direction toggle
8 Depth Set Button
9 Rotary Knobs
10 Power OFF button
11 Motor coupler
12 Threaded Shafts
13 Motor shaft bearing
14 Rotary Knobs
15 Output to locking solenoid
16 Output to Drive motor
17 Motor driver IC
18 5v Voltage regulator IC
19 Microcontroller
20 SMPS
21 Grounded casing ofvSMPS
22 Motor with gearbox
23 Engaging mechanism
24 Rotary Encoder
25 Pulley

EXTENDING MECHANISM



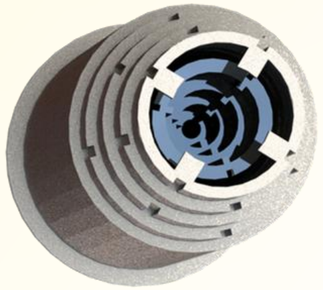


EXTENDING MECHANISM



Extending Mechanism

- The extending mechanism uses a lead screw and 3 threaded outer shafts which can slide on the lead screw and still transmit the rotation to the outer telescopic tubes which are arrested of their rotation mechanism.
- Extending: The lead screw rotates freely inside the outer shaft which is locked using a linear actuator.
- The threaded telescopic tubes ride the threads as the rotation of inner telescopic tubes is arrested by the slots with the outer tubes, but are free to move axially, thus imparting a linear motion.
- This motion co-occurs in 4 tubes and thus the mechanism can extend from 1m to 5m in 100 rotations of the lead screw.



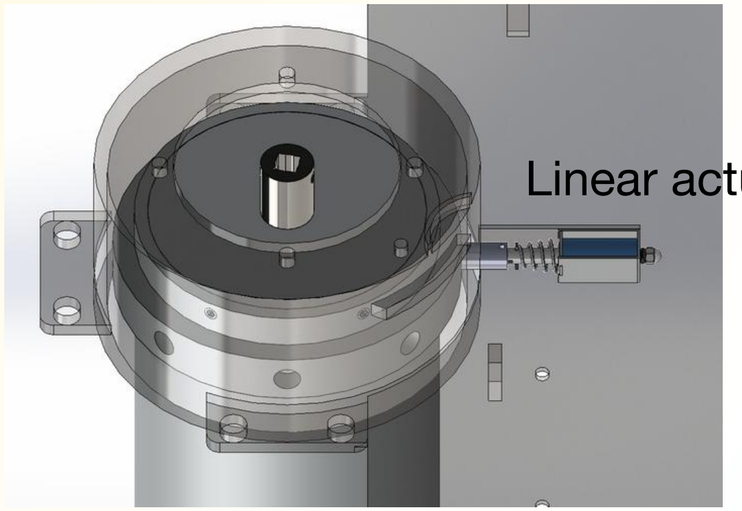
Depth Capability

- The shaft can extend from a length of 1m to 5m in 100 rotations of the lead screw.
- Full extension from 1m to 5m in 50 sec at 120RPM



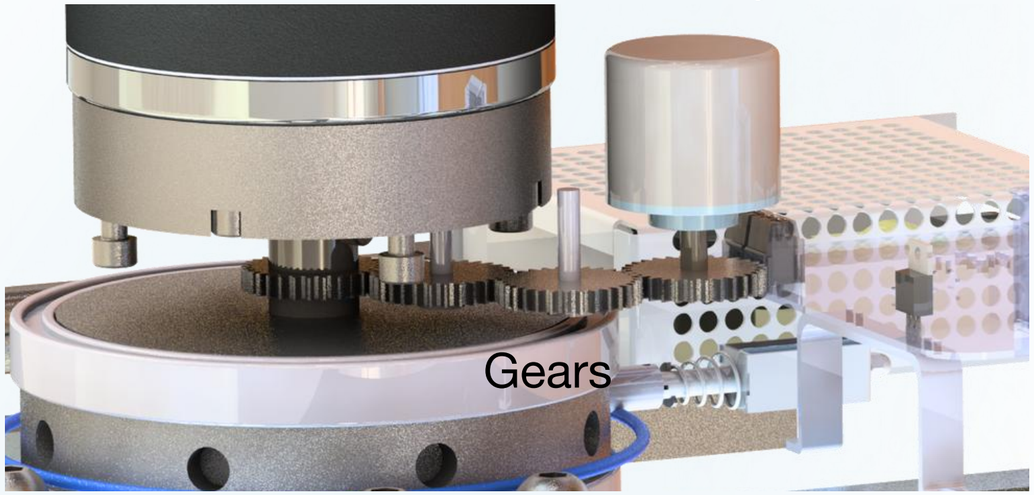
ENGAGING MECHANISM

- For operating at arbitrary depths. The rotary encoder reads the number of rotations of the screw and calculates the depth reached.
- When the desired depth is reached the linear actuator unlocks the telescopic shaft to let the impeller rotate, thus opening the blades and starting the agitation process.



Linear actuator

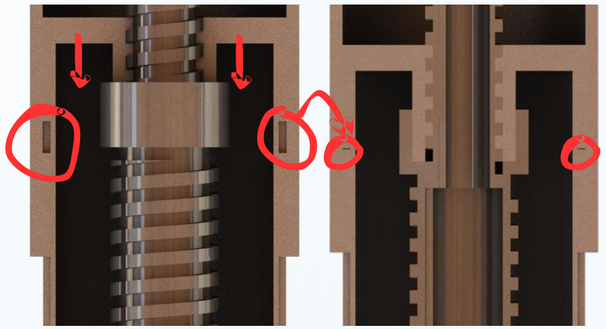
Rotary Encoder



Gears

Self-Engaging mechanism at maximum Length

- At maximum depth i.e. the slots on the telescopic tubes slide out to start the agitation mechanism automatically.
- Complete mechanical unlocking.



(At max length)

Why choose screw threaded mechanism over other alternatives ?

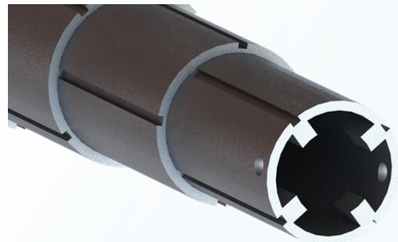
- Acts as the structural member and extending mechanism
- A single motor can be used for extending and mixing
- Complete mechanical system, avoiding critical electrical components to have direct contact with liquids.
- Self-locking of screws to rotate the impellers
- Impellers rotate only when a desired depth is reached



FLEXIBILITY AND WEIGHT

Flexibilty in operation

- The mechanism can work at **any arbitrary depth.**
- The folding blade makes it usable in **openings of different sizes**.
- Blades for specific fluids can be operated and added easily.
- Can work with slurries of varying viscosities and toxicities.



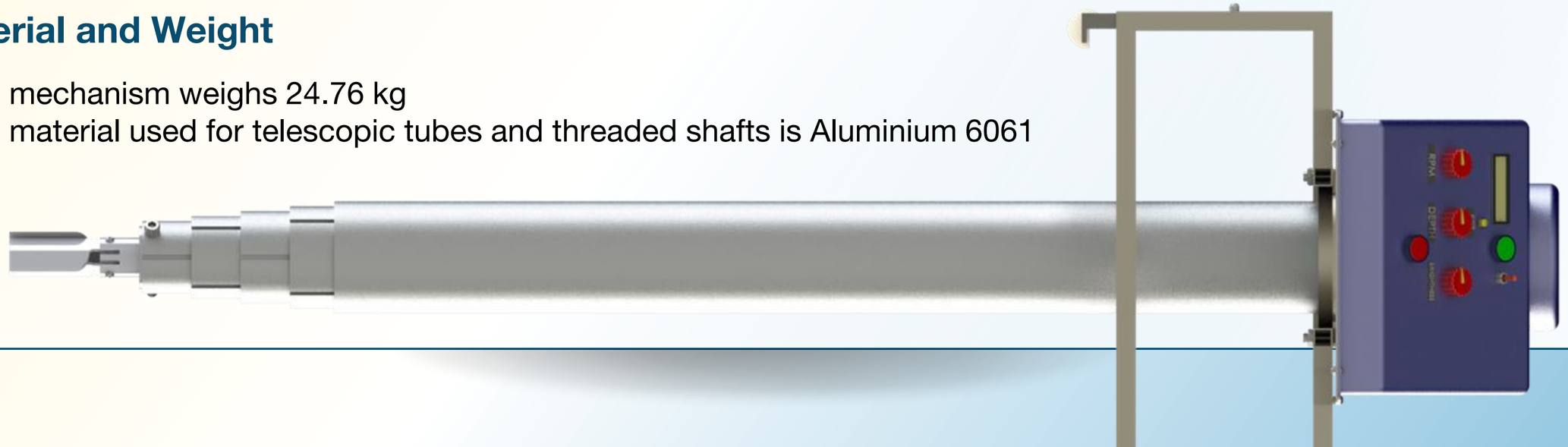


Rigidity of Shaft

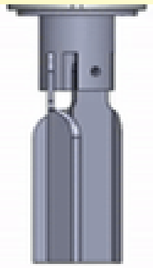
- Any bending in the shaft may cause damage to the equipment and the walls while operating and thus is undesirable.
- The threaded rods act as the structural member and thus provide rigidity for any large flexure.

Material and Weight

- The mechanism weighs 24.76 kg
- The material used for telescopic tubes and threaded shafts is Aluminium 6061



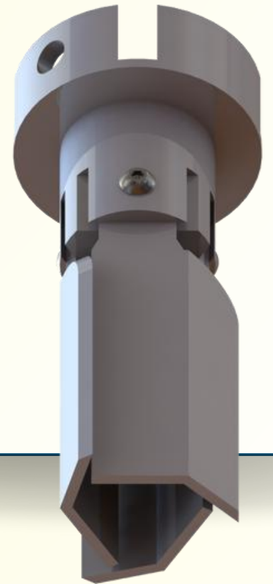
CENTRIFUGAL BLADES

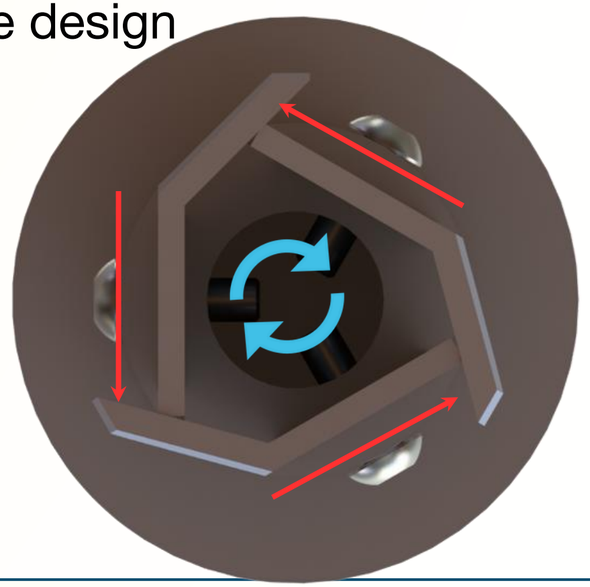


- No separate electrical components for unfolding
- Only opens when the blade starts to rotate
- Allows for use in tanks with small openings
- Closes during retraction

Changeable blades

- Single individual blades can be changed if damaged
- Impellers of different designs for specific use cases can be used due to simple swappable design





OPTIMAL BLADE DESIGN

Based on the literature study, we decided the length and thickness of the blade to be 160mm and 3mm respectively. (Ref 1)

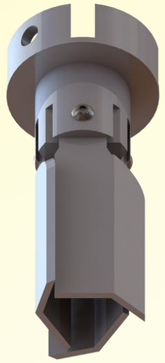
$$\dot{m} = 1181\left(\frac{d}{D}\right)^{-0.0226}$$

&

$$\dot{m} = \frac{a\left(\frac{W}{D}\right)}{1 + b\left(\frac{W}{D}\right)}$$

m= mass flow rate
d= shaft diameter = 0.02 m
D= impeller diameter = 0.16 m
a= 20,000 kg/s **(Ref 2)**
b=12
W= width of blade

W = 38 mm or 0.038 m



Close d



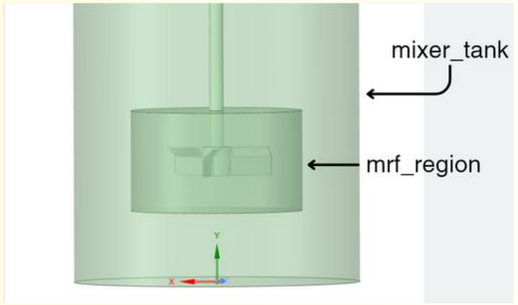
just when rotation start, blade opening



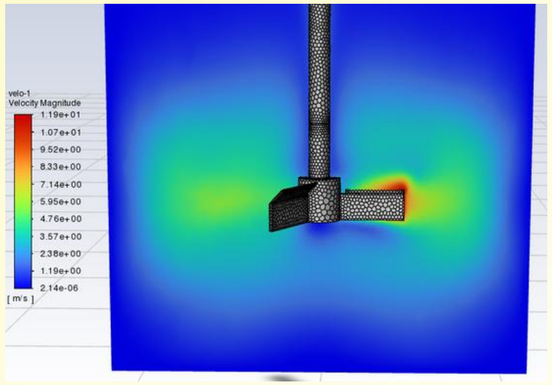
Fully open

CONFIRMED VIA ANSYS CFD SIMULATION

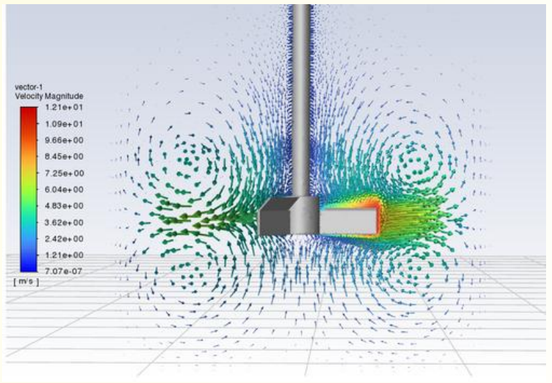
- For CFD, the Moving Reference Frame(MRF) approach is followed. The shown mrf region is rotated around the body axis while the body remains still during the simulation phase.
- The reference frame follows the Navier-Stokes equation with coriolis and centrifugal forces.



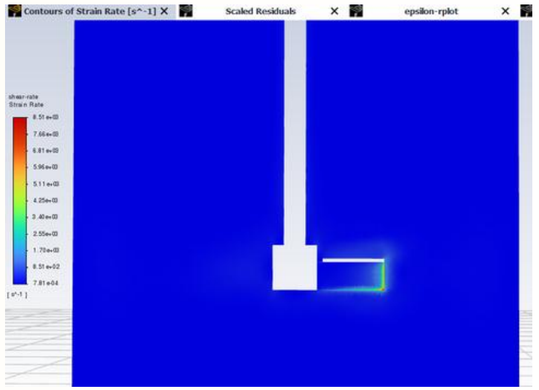
Geometry for analysis



Velocity Contour



Streamlines



Shear Rate

- From CFD results, it is found that the impeller effectively mixes the slurry within a 250 mm vertical range. To optimize mixing efficiency the impeller can be operated at different heights.

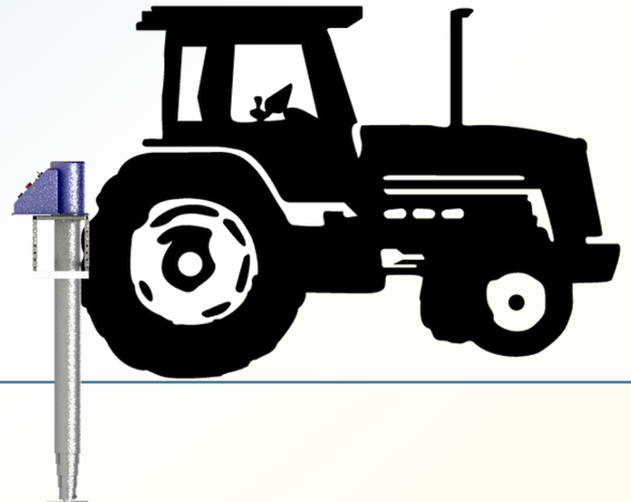
PORTABILITY

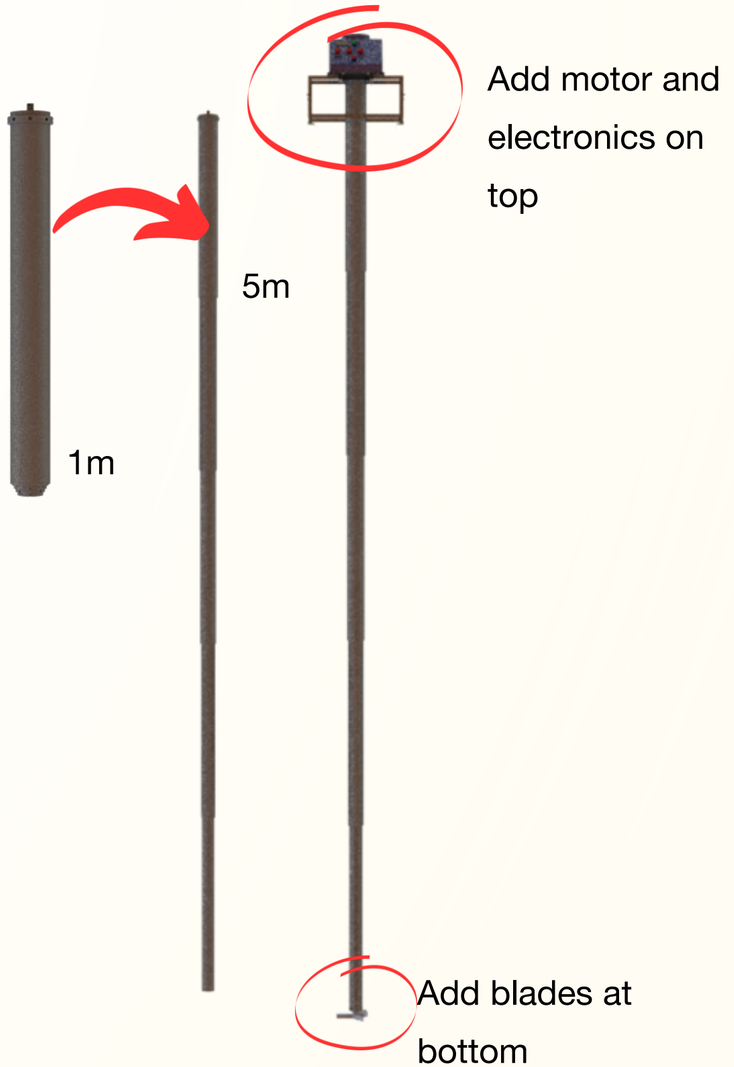
Achievable Compactness

- The mechanism weighs 24.7 Kg
- The whole assembly including the motor and impellers can be packed into a box of 1.5mx0.5mx0.5m box thus proving its compact size.

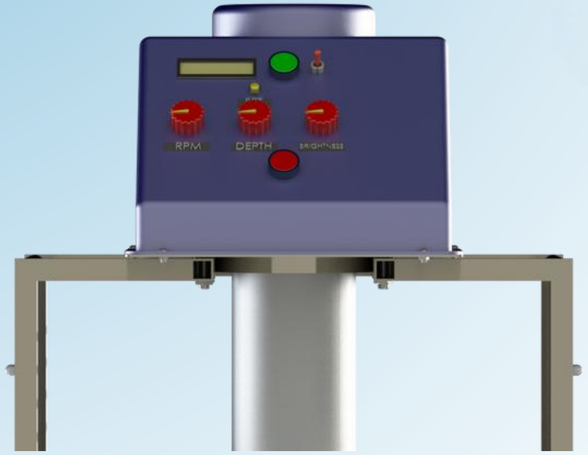
How to Maneuver it

- The stand for the whole mechanism can be built for being mounted on any kind of vehicle or kart.
- Can be mounted to vehicles and the whole mechanism can be powered by their motor for ease of use.





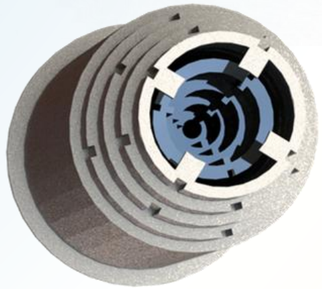
DIMENSIONS OF MACHINE



Operating Console

Width:0.28m

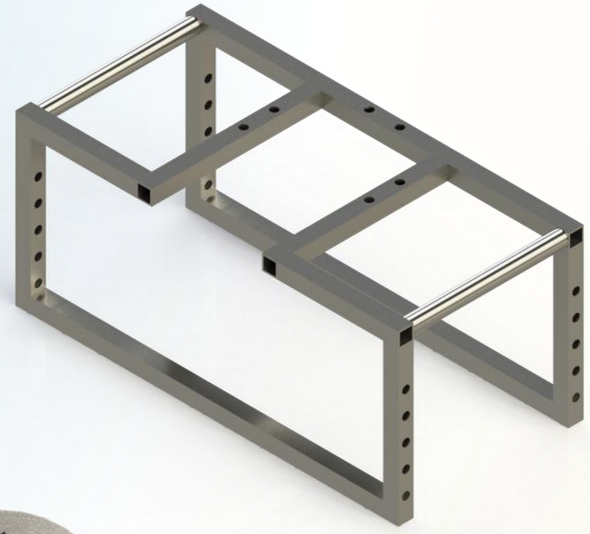
Height:0.23m



Shafts

Outer : 0.114m

Inner : 0.074m



Stand

Length: 0.5m

Width:0.24m

Height: 0.22m



Blade

Blade Diameter = 0.16 m

Blade/Tank Diameter = 0.4

Folded Blade Length = 0.13m



Bill Of Materials

S.No.	Quantity	Item	Cost in Rupees	S.No.	Quantity	Item	Cost in Rupees
1	1	100P	2	14	1	CONN-TH_2P-P5.00_DB128L-	15
2	1	DISPLAY-LCD-16X2	100	15	5	Aluminium Telescopic Tubing	11000
3	8	10k	8	16	4	Threaded Tubing	3500
4	1	SS13D07VG2	10	17	1	Impeller coupling and blades	1200
5	1	SN74HC244DBR	50	18	1	Stand	800
6	2	BTN8982TAAUMA1	100	19	1	Base	1600
7	1	LM7805T	10	20	1	Motor	5000
8	1	ARDUINO_NANO	220	21	1	Gearbox	2000
9	3	DB128L-5.0-2P-GY-S	45	22	LS	Nut and Bolts	400
10	3	SW-TH_EC11G1560421	120	23	2	Bearings	1200
11	2	CONN-TH_2P-P10.00_JL271R-10002G01	60	24	1	Washers	250
12	1	SW-TH_1AD1T1B4M2QES	10	25		***Overhead***	12000
13	1	TO-220-3_L9.9-W4.5-P2.54-L	30				
			Total =	**39730**			

Total Approximate Cost : Rs.39730

USER FRIENDLY INTERFACE

HOW TO USE

Operation

Step 1: Push slider from 'off' to 'set_depth' state.

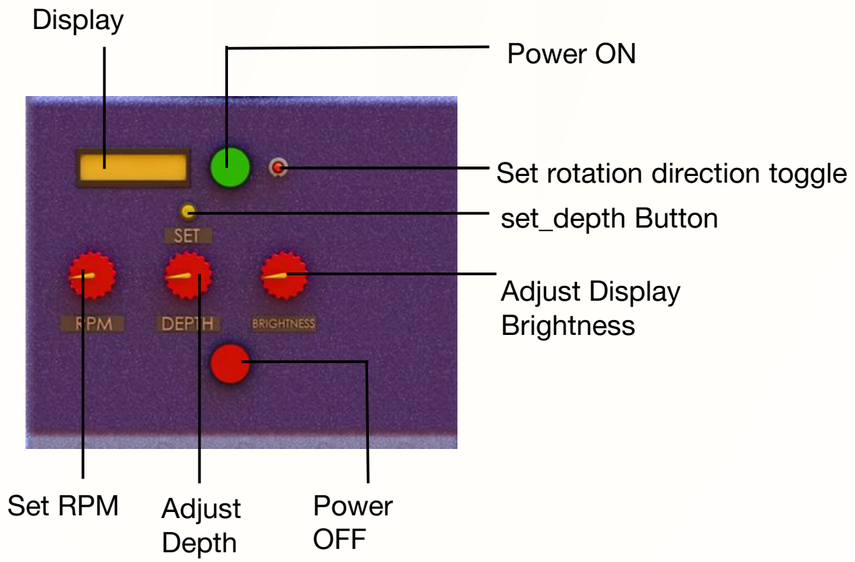
Step 2: Set depth by adjusting the depth knob, (you can see the input(desired) depth

and current depth attained on the screen)

Step 3: Once desired depth is attained, push the slider from 'set_depth' state to 'on'

state. Set direction of rotation using toggle switch.

(you can now see the depth and RPM on the screen).



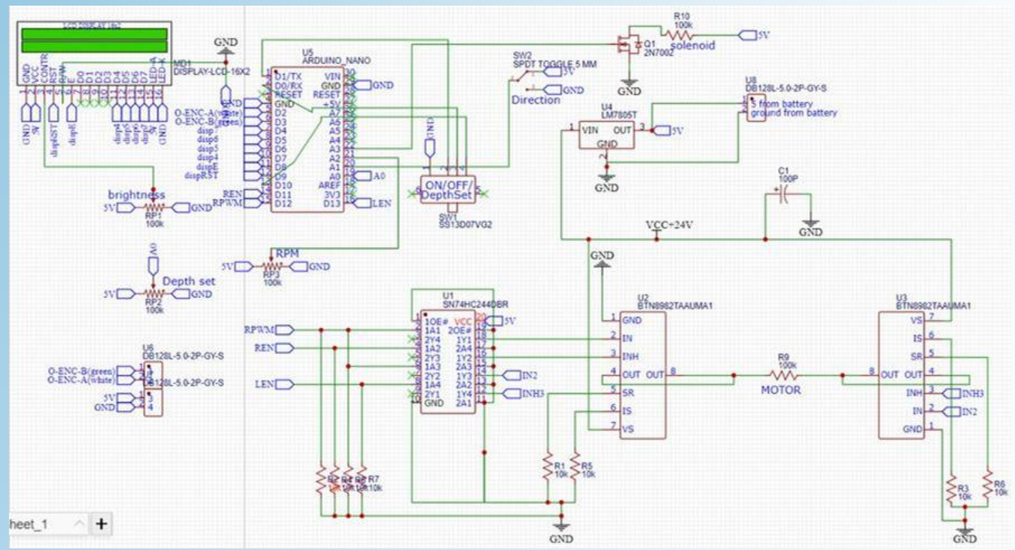
Turning off the device

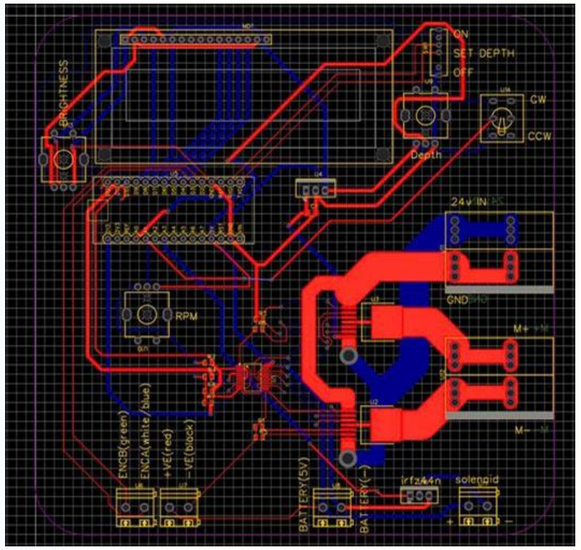
Step 1: Set RPM to 0.

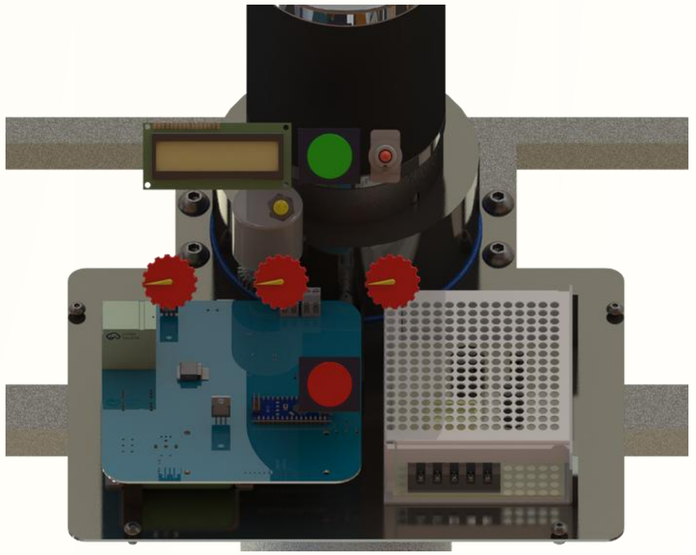
Step 2: Push slider from 'on' to 'set_depth' state, and set depth to 0.

Step 3: Push slider from 'set_depth' to 'off' state.

PCB DESIGN AND ELECTRONICS

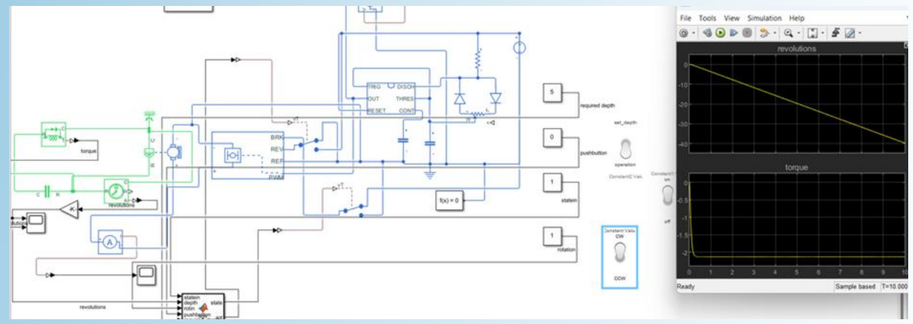




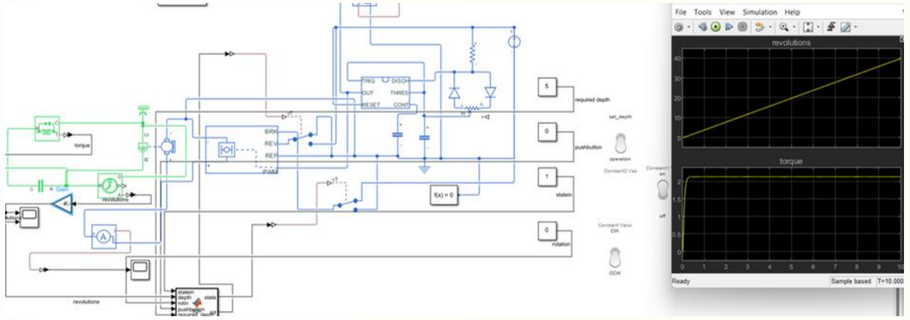


Note: that in the design, we have used 2 switches to select any of the 3 modes, whereas in the circuit and code we have used a single switch with 3 states. Any one of the following can be implemented int he final design, based preference

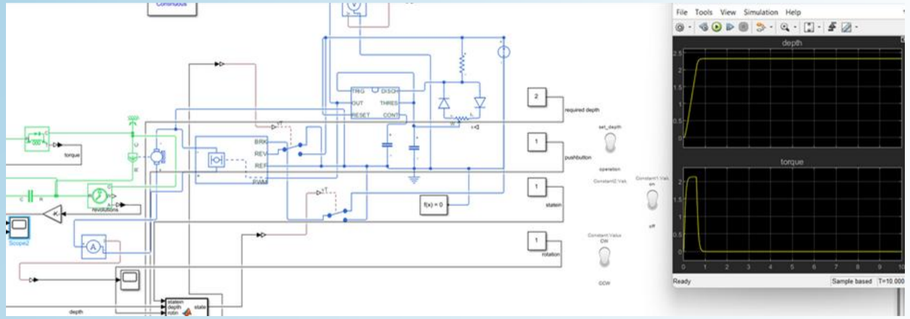
SIMULINK REPRESENTION



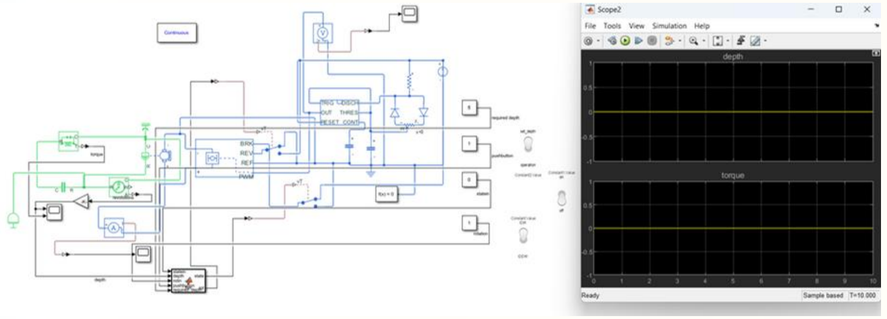
(1) operation mode (clockwise)



(2) operation mode (counter clockwise)



(3) depth set to 2m.



(4) off state

the Simulink diagram shows 4 different operational states.

(1) when on state in enabled and toggle is flipped to clockwise - we see a negative Rotation of the whole shaft, and torque for the whole time

(2) we can see the same thing but in the opposite direction

(3) depth has been given as 2m, so the shaft will extend up to 2m depth. note that whenever depth input is changed, the shaft will attain that depth only in the 'set_depth' mode, and will cu toff power to the motor upon reaching the desired depth.

(4) off state: cuts off power to the motors, and locks the shaft so that its length doesn't change

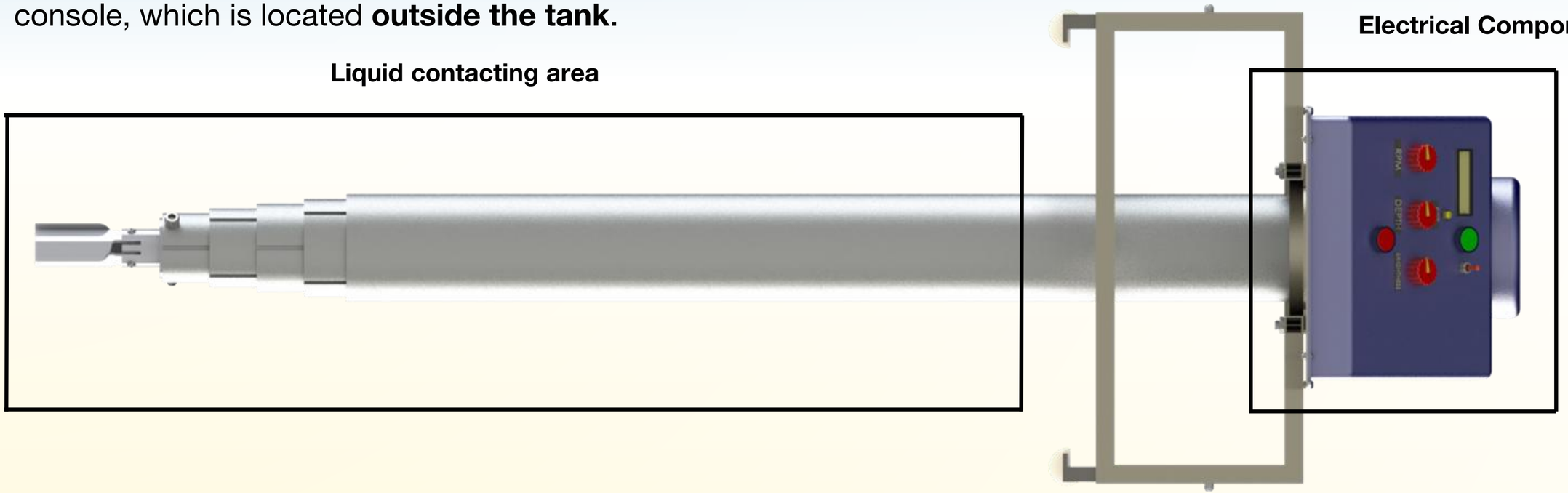
WATERPROOFING

Why our device is completely Waterproof?

- The most effective approach to waterproofing is to eliminate the need for waterproofing altogether by devising a design that prevents direct exposure of electrical components to liquid-prone areas within machines.
- In the context of this mechanism, **the extending mechanism is entirely mechanical, ensuring that all electrical components are positioned** on the central operating console, which is located **outside the tank.**

Liquid contacting area

Electrical Components



How to deal with cavitation and corrosion?

- Our device uses wear-resistant coating such as Metaline to prevent cavitation
- The walls of the extending telescoping shaft are quite thick to prevent shaft damage in the long run
- Risk is mainly in the blades which can easily be replaced with minimum cost. see blade changing mechanism.
- We can also use bionic propellers to reduce cavitation (Con- Tough geometry to manufacture)
- Use anti-vibration mounts to reduce mechanical stress on seals and joints.

Additional Measures

- Pre-testing and regular maintenance - Perform rigorous testing under various water exposure scenarios beforehand and while deployed check for regular care
- Smooth Surface Finish: Ensure a smooth surface finish on all components to minimize water retention.
- Ingress Protection (IP) rating Compliance: Aim for an IP rating that aligns with your device's usage environment. IP ratings specify the level of protection against solids and liquids, providing a standardized measure.
- Hydrophobic Materials: These materials naturally repel water, minimizing the chances of water adherence to surfaces.

DESIGN SAFETY

DE GOODMAN CRITERIA

We have selected the innermost rotating threaded shaft (which is coupled to the motor) as a Critical Region for failure. When the shaft is in maximum extension i.e. 5m, then the shaft will be most likely to fail.

So considering that, There will be an axial Force Fa (Due to mass of entire mechanism, “mg”)=(24,78 kgX9.8)Newton and torque = 350 Nm

Determining Fatigue Factor of Safety by DE Goodman criteria

(A) For the calculation of standard alternating and midrange stresses

$$\sigma_a = K_f \frac{M_a c}{I} \qquad \sigma_m = K_f \frac{M_m c}{I}$$

$$\tau_a = K_{fs} \frac{T_a c}{J} \qquad \tau_m = K_{fs} \frac{T_m c}{J}$$

$$\sigma_a = K_f \frac{32 M_a}{\pi d^3} \qquad \sigma_m = K_f \frac{32 M_m}{\pi d^3}$$

$$\tau_a = K_{fs} \frac{16 T_a}{\pi d^3} \qquad \tau_m = K_{fs} \frac{16 T_m}{\pi d^3}$$

(B) For calculating von-mises stress

$$\sigma'_a = (\sigma_a^2 + 3\tau_a^2)^{1/2} = \left[\left(\frac{32 K_f M_a}{\pi d^3} \right)^2 + 3 \left(\frac{16 K_{fs} T_a}{\pi d^3} \right)^2 \right]^{1/2}$$

$$\sigma'_m = (\sigma_m^2 + 3\tau_m^2)^{1/2} = \left[\left(\frac{32 K_f M_m}{\pi d^3} \right)^2 + 3 \left(\frac{16 K_{fs} T_m}{\pi d^3} \right)^2 \right]^{1/2}$$

(C) Failure Criterian - Modified Goodman Theory

$$\frac{1}{n} = \frac{\sigma'_a}{S_e} + \frac{\sigma_m}{S_{ut}}$$

$$\frac{1}{n} = \frac{16}{\pi d^3} \left\{ \frac{1}{S_e} \left[4(K_f M_a)^2 + 3(K_{fs} T_a)^2 \right]^{1/2} + \frac{1}{S_{ut}} \left[4(K_f M_m)^2 + 3(K_{fs} T_m)^2 \right]^{1/2} \right\}$$

In or case for Al 6061

Sut =310 Mpa

Sy = 260 MPa

d = 20mm

J = 1.5708 cm^4

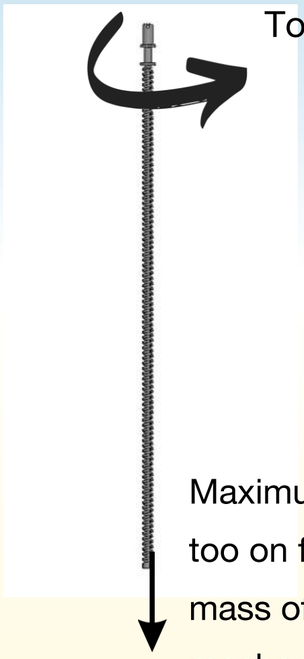
I = 0.7854 cm^4

Kt = 2.09

v = 0.33

$$K_f = 1 + q(K_t - 1) \quad \text{or} \quad K_{fs} = 1 + q_{\text{shear}}(K_{ts} - 1)$$

q= notch sensitvity



Maximum Axial load that too on full extension (= mass of the entire mechanism)

(Critical breaking region FBD)

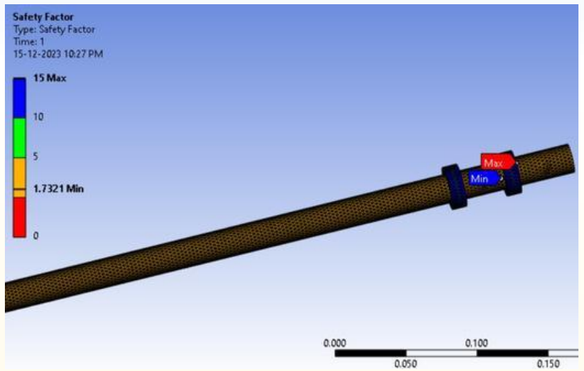
Factor of safety - 1.84 Theoretical

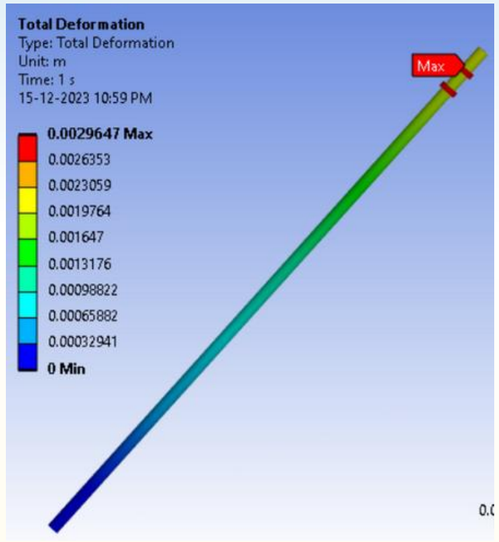
FEA SIMULATION RESULTS

Confirmed via FEA simulation

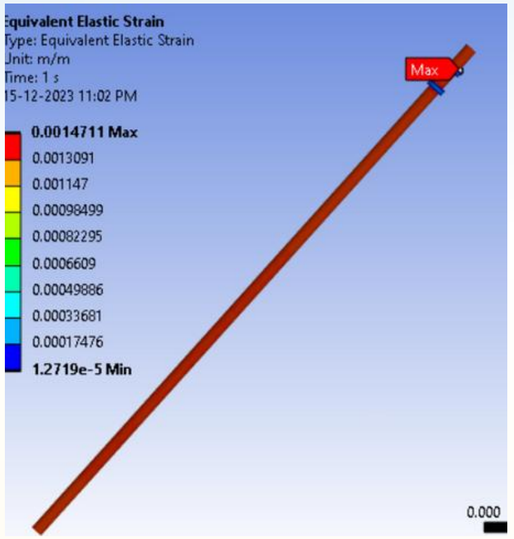
Material Selected - Al 6061, T6

Ultimate tensile strength, Sy = 260 MPa





Total Deformation



Equivalent Elastic Strain

Factor of safety - 1.7321 (at most critical point)

BEARING SELECTION

Bearing Selection via SKF catalogue

$$F_e = (XVF_r + YF_a)C_s$$

$$\left(\frac{C}{F_e}\right)^a = L$$

Here Fa = 24.78 kg or Fa = 242.84 N Bore dia = internal dia of threaded shaft = 20mm

Here Fa = 24.78 kg or Fa = 242.84 N

Co = 5000, Cs =1 , C= 9950 , a=3 (deep groove ball bearing)

$\frac{F_a}{C_o}$ = 0.048 = > e = 0.24 = > x= 0.53 and y = 1.85 = > Fe = 449.25

= > L10 = 10864.38 million revolution (Rating Life)

Selected Bearing - 6004 (Deep groove ball bearing, SKF catalogue)

REFERENCES

- https://www.youtube.com/watch?v=kobc3DdaofE&t=52s
- Jimmy Lea, Suspension Mixing Tank- Design Heuristic, Chemical Product and Process Modeling, Vol 4, Issue 1, 2007, Article 17
- Sumit R.Desai, S.Y.Gajjal and Kulloli, Redesign and Structural analysis of agitator shaft for reactor pressure vessel, International Journal of Current Engineering and Technology, 2016
- Shigley's Mechanical Engineering Design
- circuit for operating the motor driver IC: https://community.simplefoc.com/t/3x-bts7960-design-for-bldc/2387
- Lead Screw Telescopic Mechanism: https://www.researchgate.net/figure/Structure-diagram-of-lead-screw-and-telescopic-sleeve_fig2_374041957
- https://metaline.com/en/applications/petrochemical-oil-gas/agitators-mixers.html
- https://bigrep.com/wp-content/uploads/2018/03/Bionic-Propeller_eBook.pdf
- https://www.linak.com/segments/techline/tech-trends/ingress-protection/

Thank You